Transfer task: defend, revise, submit

LEARNING TARGET I can explain my prediction, identify limits, and improve my reasoning.

A · KEY VOCABULARY

Claim	Your prediction, stated clearly.
Evidence	Visible work: a graph, sketch, or calculation.
Limitation	What the model leaves out or assumes.
Revision	A corrected step, shown — and how it changes later results.

B · KEY HABIT

`a correct-looking number ≠ shown reasoning`

`physics predicts; people decide`

`central value` + its range
`revision` show the fixed step

C · COLOR IT · BUILD A DEFENSE

Point to your own work as you speak.

1 CLAIM
I predict ___ .

2 EVIDENCE
based on ___ (point to the graph, sketch, or calculation).

3 REASONING
because my model assumes ___ .

4 LIMITATION
The result is limited by ___ . A range is not a probability, and a recommendation also involves cost, feasibility, and priorities.

- ☐ **1 · BLUE** Claim: your predicted position
- ☐ **2 · CODE COLOR** Evidence: color by quantity
- ☐ **3 · PENCIL** Reasoning
- ☐ **4 · PENCIL** Limitation

D · WORKED EXAMPLE

Defend a prediction: a 2 kg cart under 4 N net force, from rest at 1 m, for 2 s; model bound ± 1 m.

GIVEN	EQUATION	WORK	ANSWER
F_{net} = 4 N, m = 2 kg x_i = 1 m, t = 2 s bound ± 1 m	force → a → v → x	a = 2 m/s², v = 4 m/s Δx = 4 m, x_f = 5 m	**4 to 6 m** assumes constant forces

E · THE 2026-XJ THREAD

DAY 22 OF 22 · NEW TOOL: YOUR DEFENSE

The desk wants a claim it can defend: where XJ will be, how sure you are, and what your model leaves out. That is what makes a prediction trustworthy.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Explain what the forecast means and what limits it

Claim, evidence, reasoning, and a visible revision

A correct-looking number is not the same as shown reasoning. Someone reading a forecast needs to know what was predicted, which evidence supports it, and what assumptions connect the two. Today I want the explanation to be clear enough that another person could follow the calculation and understand its limits.

A claim states the result. Evidence is the visible work supporting it: measurements, a graph, a sketch, or a calculation. Reasoning explains why the physics model connects that evidence to the claim. A limitation identifies something uncertain, assumed, or left out. These four parts work together rather than replacing one another.

The worked example uses a **2 kg cart** under a constant **4 N net force**. It starts at rest at **1 m**, and we predict its position after **2 s**. First find $a = 4\ \text{N} \div 2\ \text{kg} = 2\ \text{m/s}^2$. Then $v_f = 0\ \text{m/s} + (2\ \text{m/s}^2)(2\ \text{s}) = 4\ \text{m/s}$. The constant net force and fixed mass justify constant acceleration.

Average velocity is $(0 + 4)\ \text{m/s} \div 2 = 2\ \text{m/s}$, so $\Delta x = (2\ \text{m/s})(2\ \text{s}) = 4\ \text{m}$. Add the starting coordinate: $x_f = 1\ \text{m} + 4\ \text{m} = 5\ \text{m}$. The example supplies a model bound of $\pm 1\ \text{m}$, so the reported position interval is **4 m to 6 m**. The bound is part of the supplied information, not something this arithmetic creates by itself.

A defense can now connect the pieces: we predict a central position of 5 m after two seconds, with the given range from 4 m to 6 m. The evidence is the force, mass, starting conditions, and time. The reasoning uses Newton's second law and constant-acceleration motion. The result is limited by the supplied bound and the assumption that the forces stay constant throughout the interval.

A range is not a probability. This example does not tell us that some percentage of outcomes must occur inside it. Also, a model can miss a changing force even when its arithmetic is correct. I want uncertainty in the inputs and limits of the model named honestly, rather than hidden behind extra decimal places.

A revision should show the corrected step and its effect on later results. For example, if the starting coordinate were corrected from 1 m to 2 m, with everything else unchanged, the displacement would still be 4 m but the central final position would become $2\ \text{m} + 4\ \text{m} = 6\ \text{m}$. We can explain exactly why that result changed instead of simply replacing the old answer.

For our fictional asteroid, 2026-XJ, a useful forecast includes a location, future time, reference frame, supporting work, and limits. Decisions about a response also involve cost, feasibility, and priorities. Physics supplies evidence for those decisions. A clear defense shows what the model supports and what additional information people still need.